Double- and Half-Angle Identities

Double-Angle Identities

In the last lesson, we applied the sum and difference identities for sine, cosine, and tangent.

$$\sin(A+B) = \sin(A)\cos(B) + \cos(A)\sin(B)$$

$$\cos(A+B) = \cos(A)\cos(B) - \sin(A)\sin(B)$$

$$\tan(A+B) = \frac{\tan(A)+\tan(B)}{1-\tan(A)\tan(B)}$$

In the case where $A = B$, we have the following…

$\sin(A+A) = \sin(2A)$ $\quad$ $\cos(A+A) = \cos(2A)$ $\quad$ $\tan(A+A) = \tan(2A)$

While the angle sum identities still apply here, generally, we refer to these as double angles and use the following formulas.

PROPERTY: ***The Double-Angle Identities***
Let θ be any value for which the following functions are defined.
$$\sin(2\theta) = 2\sin(\theta)\cos(\theta)$$ $$\cos(2\theta) = \cos^2(\theta) - \sin^2(\theta)$$ $$\tan(2\theta) = \frac{2\tan(\theta)}{1-\tan^2(\theta)}$$

The double-angle identities above are the most common. However, there are some useful variations of the double-angle identity for cosine.

EXAMPLE 1

Use the double-angle identities above, along with other fundamental identities, to verify the following variations of the double-angle identity for cosine.

$\cos(2\theta) = 1 - 2\sin^2(\theta)$

$\cos^2\theta - \sin^2\theta$ ✓

$(1 - \sin^2\theta) - \sin^2\theta$

$\cos^2\theta - \sin^2\theta$ ✓

$\cos(2\theta) = 2\cos^2(\theta) - 1$

$\cos^2\theta - \sin^2\theta$ ✓

$\cos^2\theta + (\cos^2\theta - 1)$

$\cos^2\theta - (1 - \cos^2\theta)$

$\cos^2\theta - \sin^2\theta$ ✓

We've seen before that given the sine, cosine, or tangent value for an angle θ, along with the quadrant to which θ belongs, we can evaluate any of the other trigonometric functions for the angle. Equipped with the double-angle identities, angle sum/difference identities, and even/odd identities, ***we can extend this to <u>any</u> integer multiple of the angle θ.***

EXAMPLE 2

Let $\cos(\theta) = -4/5$ with $\theta \in QIII$.

(a) Find $\sin(2\theta)$, $\cos(2\theta)$, and $\tan(2\theta)$.

$\sin^2\theta = 1 - \cos^2\theta$

$\sin\theta = \pm\sqrt{1-(-4/5)^2}$

$= -\sqrt{1-16/25}$

$= -\sqrt{9/25}$

$= -\frac{3}{5}$

$\sin(2\theta) = 2\sin\theta\cos\theta = 2\left(-\frac{3}{5}\right)\left(-\frac{4}{5}\right) = \frac{24}{25}$

$\cos(2\theta) = \cos^2\theta - \sin^2\theta = \left(-\frac{4}{5}\right)^2 - \left(-\frac{3}{5}\right)^2 = \frac{16}{25} - \frac{9}{25} = \frac{7}{25}$

$\tan(2\theta) = \frac{\sin(2\theta)}{\cos(2\theta)} = \frac{24/25}{7/25} = \frac{24}{7}$

(b) Find $\sin(4\theta)$ and $\sin(-4\theta)$.

$\sin(2\theta) = \frac{24}{25}$

$\cos(2\theta) = \frac{7}{25}$

$\sin(4\theta) = \sin(2\cdot 2\theta) = 2\sin(2\theta)\cos(2\theta)$

$= 2\left(\frac{24}{25}\right)\left(\frac{7}{25}\right) = \frac{336}{625}$

$\sin(-4\theta) = -\sin(4\theta) = -\frac{336}{625}$

(c) Find $\cos(3\theta)$ and $\cos(-3\theta)$.

$\sin(\theta) = -\frac{3}{5}$

$\cos(\theta) = -\frac{4}{5}$

$\sin(2\theta) = \frac{24}{25}$

$\cos(2\theta) = \frac{7}{25}$

$\cos(3\theta) = \cos(2\theta+\theta) = \cos(2\theta)\cos(\theta) - \sin(2\theta)\sin(\theta)$

$= \frac{7}{25}\left(-\frac{4}{5}\right) - \frac{24}{25}\left(-\frac{3}{5}\right) = -\frac{28}{125} + \frac{72}{125} = \frac{44}{125}$

$\cos(-3\theta) = \cos(3\theta) = \frac{44}{125}$

In calculus, the double-angle identities play an essential role when integrating powers of sine and cosine. In particular, they are often used as "power-reducing" identities. Therefore, you should take a moment to derive the power-reducing identities below for yourself.

$$\cos^2(\theta) = \frac{1}{2}(1+\cos(2\theta)) \qquad \sin^2(\theta) = \frac{1}{2}(1-\cos(2\theta))$$

Half-Angle Identities

Much like the double-angle identities are a direct consequence of the angle sum identities, the half-angle identities result from the double-angle identities.

PROPERTY: *The Half-Angle Identities*

Let θ be any value for which the following functions are defined. Note the $\pm$ sign here is determined by the quadrant for the angle $\theta/2$.

$$\sin\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1-\cos(\theta)}{2}} \qquad \cos\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1+\cos(\theta)}{2}} \qquad \tan\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1-\cos(\theta)}{1+\cos(\theta)}}$$

A WORD OF CAUTION: Some students mistakenly leave the ± sign instead of selecting an appropriate sign when evaluating the above statements. Thinking back to your experience in an algebra course, why is this not possible? **Hint:** these are trigonometric ***functions***. Recall vertical line test! One output for each input.

As mentioned above, these identities are a direct result of the double-angle identities.

EXAMPLE 3

Starting with the double-angle identity, $\cos(2x) = 1 - 2\sin^2(x)$, derive the half-angle identity for sine.

$\cos(2x) = 1 - 2\sin^2(x)$

$2\sin^2(x) = 1 - \cos(2x)$

$\sin^2(x) = \dfrac{1-\cos(2x)}{2}$

$\sin(x) = \pm\sqrt{\dfrac{1-\cos(2x)}{2}}$

Let $x = \dfrac{\theta}{2}$

$\sin\left(\dfrac{\theta}{2}\right) = \pm\sqrt{\dfrac{1-\cos(2\cdot\theta/2)}{2}}$

$\sin\left(\dfrac{\theta}{2}\right) = \pm\sqrt{\dfrac{1-\cos(\theta)}{2}}$

Evaluating Trigonometric Expressions

Like the angle sum/difference identities, the double- and half-angle identities allow us to evaluate trigonometric functions for a greater variety of angle measures.

EXAMPLE 5

It is well known that $\sin(90°) = 1$ and $\sin(45°) = \cos(45°) = \sqrt{2}/2$. Confirm the double- and half-angle identities will yield these expected results.

$\sin(90°) = \sin(2 \cdot 45°) =$ $2\sin(45°)\cos(45°)$

$= 2\left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{2}}{2}\right)$

$= 2\left(\frac{2}{4}\right) = 1$

$\sin(45°) = \sin\left(\frac{90°}{2}\right) = +\sqrt{\frac{1-\cos(90°)}{2}}$, $\sin(45°) > 0$

$= \sqrt{\frac{1-0}{2}}$

$= \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$ ✓

The half-angle identities outlined so far are the most common. However, there are useful variations of the double-angle identity for tangent which are worth investigating.

$$\tan\left(\frac{\theta}{2}\right) = \frac{\sin(\theta)}{1+\cos(\theta)} \qquad \tan\left(\frac{\theta}{2}\right) = \frac{1-\cos(\theta)}{\sin(\theta)}$$

EXAMPLE 5

Evaluate $\tan\left(\frac{\pi}{8}\right)$.

$\tan\left(\frac{\pi}{8}\right) = \tan\left(\frac{\pi/4}{2}\right) = \frac{\sin(\pi/4)}{1+\cos(\pi/4)} = \frac{\sqrt{2}/2}{1+\sqrt{2}/2}\left(\frac{2}{2}\right)$

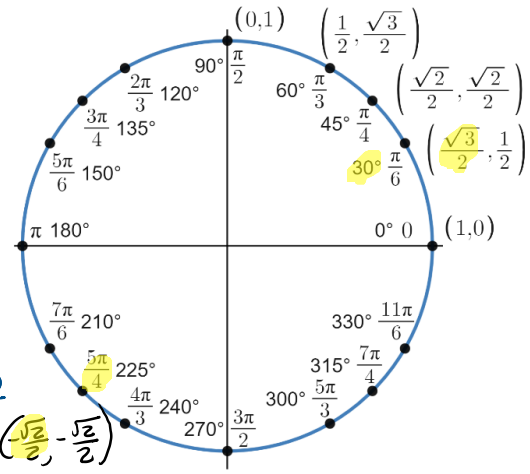
$= \frac{\sqrt{2}}{2+\sqrt{2}}\left(\frac{2-\sqrt{2}}{2-\sqrt{2}}\right)$

$= \frac{2\sqrt{2}-2}{4-2\sqrt{2}+2\sqrt{2}-2}$

$= \frac{2(\sqrt{2}-1)}{2} = \sqrt{2}-1$

EXAMPLE 6

Evaluate each expression.



(a) $\sin(15°) = \sin\left(\frac{30°}{2}\right) = +\sqrt{\frac{1-\cos(30°)}{2}}$ ($15° \in$ QI)

$= \sqrt{\frac{1-\sqrt{3}/2}{2}\left(\frac{2}{2}\right)} = \sqrt{\frac{2-\sqrt{3}}{4}}$

$= \frac{\sqrt{2-\sqrt{3}}}{2}$

(b) $\cos\left(\frac{5\pi}{8}\right) = \cos\left(\frac{5\pi/4}{2}\right) = -\sqrt{\frac{1+\cos(5\pi/4)}{2}}$ ($\frac{5\pi}{8} \in$ QII)

$= -\sqrt{\frac{1-\sqrt{2}/2}{2}\left(\frac{2}{2}\right)} = -\sqrt{\frac{2-\sqrt{2}}{4}}$

$= -\frac{\sqrt{2-\sqrt{2}}}{2}$

EXAMPLE 7

(a) Evaluate the expression $\sin(105°)\cos(105°)$, using half-angle identities.

$\sin(105°) = \sin\left(\frac{210°}{2}\right) = +\sqrt{\frac{1-\cos(210°)}{2}} = \sqrt{\frac{1-(-\sqrt{3}/2)}{2}\left(\frac{2}{2}\right)} = \sqrt{\frac{2+\sqrt{3}}{4}} = \frac{\sqrt{2+\sqrt{3}}}{2}$

($105° \in$ QII)

$\cos(105°) = \cos\left(\frac{210°}{2}\right) = -\sqrt{\frac{1+\cos(210°)}{2}} = -\sqrt{\frac{1-\sqrt{3}/2}{2}\left(\frac{2}{2}\right)} = -\sqrt{\frac{2-\sqrt{3}}{4}} = -\frac{\sqrt{2-\sqrt{3}}}{2}$

$\sin(105°)\cos(105°) = \frac{\sqrt{2+\sqrt{3}}}{2}\left(-\frac{\sqrt{2-\sqrt{3}}}{2}\right) = -\frac{\sqrt{4-2\sqrt{3}+2\sqrt{3}-3}}{4} = -\frac{1}{4}$

(b) Evaluate the expression $\sin(105°)\cos(105°)$, using a double-angle identity.

$\sin(2\theta) = 2\sin\theta\cos\theta \rightarrow \sin\theta\cos\theta = \frac{1}{2}\sin(2\theta)$

$\sin(105°)\cos(105°) = \frac{1}{2}\sin(210°) = \frac{1}{2}\left(-\frac{1}{2}\right) = -\frac{1}{4}$